NUTRIGENOMIC DIETS: A ROUGH GUIDE

Some genes that impact nutrition and hint at various diseases, and the nutrigenomic diets that can keep them at bay

DRB2 (Beta 2 Adrenegic Receptor): Variants in DRB2 can lead to reduced carbohydrate output in cells, contributing to **Type 2 diabetes** and obesity

DIETS To avoid diabetes, diets like **Ketogenic Diet** or a **low carbohydrate diet** and fasting are popular. They keep blood sugar low to ensure insulin sensitivity doesn't fall

TAS2R38: This gene impacts how much **bitterness we taste in food.** People with this gene mutation eat less vegetables and have preference for sweets

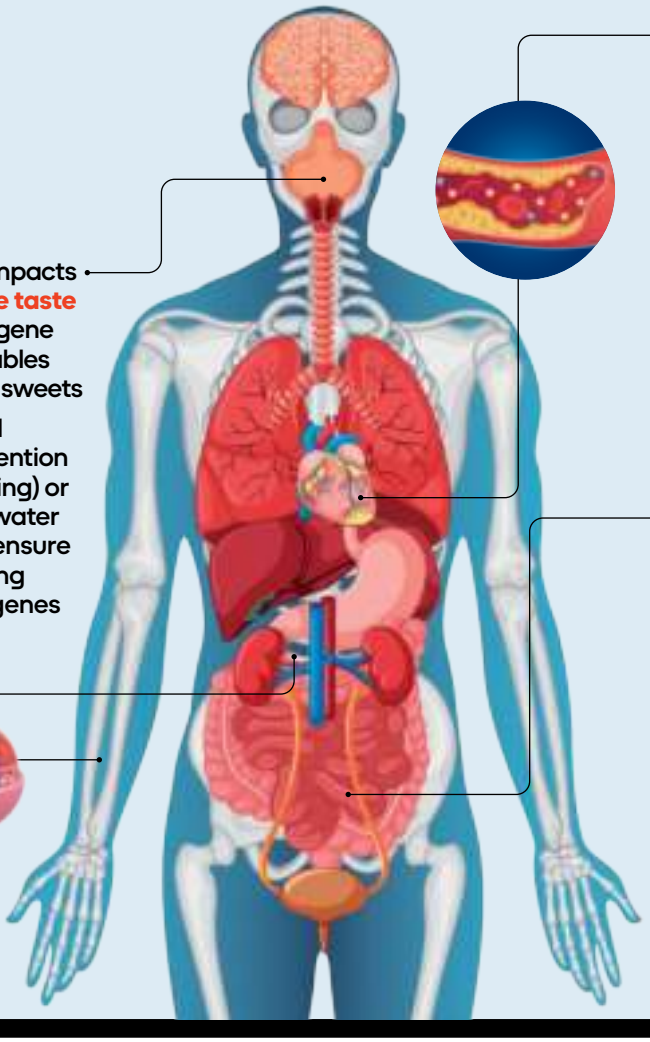
DIETS Diets like **Mindful Eating** (that calls for attention to the experience of eating) or infusing vegetable into water to dilute bitterness can ensure one doesn't eat the wrong foods goaded by one's genes



MTHFR (Methylene Tetrahydrofolate Reductase): Variations in this gene can affect folate metabolism, which provides crucial one-carbon units for **cellular functions**

DIETS To avoid folate (or Vitamin B9) deficiency that can cause anaemia and neurological issues, one can opt for a **Folate Diet**, which comprises supplements, citrus fruits and leafy vegetables





search firm IMARC, the Indian genetic testing market was valued at over $1.8 billion (Rs 15,400 crore) in 2024.

However, as Dr Aparna Bhanushali, director, medical genetics, HaystackAnalytics, points out, "While accuracy at the technical level—meaning the ability to detect genetic variants—is very high, the clinical interpretation of these variants in the context of nutrition is still an emerging area." The predictive power of nutrigenomic insights can vary depending on the complexity of gene-nutrient interactions and the strength of scientific evidence backing them, she says.

Knowing the way the body uses a nutrient through nutrigenomic testing is one part. The other half is using this information for better health. This is done not only by targeted nutrients but also by epigenetic medicine, which targets epigenetic mechanisms to modify gene expression and treat specific diseases.

"Nutrigenomics provides a personalised nutrition model grounded in molecular biology, epigenetics and clinical nutrition," says Dr Bhanushali. "For example, variations (also known as polymorphisms) in the FTO gene (a fat mass and obesity-associated gene) are associated with increased obesity risk, while variants in the MTHFR (Methylene Tetrahydrofolate Reductase) gene can impair folate (a nutrient in Vitamin B complex essential to the body) metabolism, increasing the risk of cardiovascular disease and neural tube defects."

According to market research firm Grand View Research, the Indian epigenetics market generated a revenue of $479.8 million (Rs 4,150 crore) in 2023. Health tech startup Vieroots, which raised $136,000 (Rs 1.16 crore) in funding this year and is valued at Rs 106 crore, says it provides an EPLIMO (epigenetic lifestyle modification) programme based on a geno-metabolic assessment. EPLIMO can detect multiple genetic variants causing over 250 diseases like diabetes, hypertension, cardiovascular diseases, cancers, depression etc. years or decades before they set in, enabling users to change their lifestyles. "Over 10,000 people have tried EPLIMO. We have seen amazing improvements in their health markers after three months, when we do a blood work. This proves that biohacking is measurable," says Sajeev Nair, founder and chairman, Vieroots. Nair follows a 'Sajeev diet' plan. "Once I started following this hyperpersonalised diet plan, I started seeing great results including a stable gut, higher energy levels, proper digestion and better cognitive functions," says Nair.

FUTURE OF NUTRITION

According to experts, relying on the right facts while eating what works for the body is where the future of nutrition